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BARANGGAY PROFILING SYSTEM

A System Proposal Presented to the
Faculty of the College Department
College of Computing and Information Sciences
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In Partial Fulfillment of the
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Bachelor of Science in Information Technology

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CHAPTER 1

INTRODUCTION

Barangay profiling is a system used by local government units, particularly here in the Philippines, to gather and analyze data about each barangay. The purpose of barangay profiling is to obtain a comprehensive understanding of the demographic, social, economic, and environmental characteristics of each barangay. Here's a brief overview of how the barangay profiling system typically works:

1. Create User Account:

Users begin by registering for an account on the platform. They provide essential information such as a username, email address, and password.

2. Register a Resident:

Users can register a resident within the designated barangay and obtain required information from the resident to complete the registration process.

Personal information includes:

- Full name
- Date of birth
- Gender
- Contact information.

3. View Resident List:

Users can view, update, and delete the persons they have registered.

Overview

The Barangay Profiling System is suitable for usage by departmental staff who require this information for their business unit as well as barangay employees who have access to profile information of barangay residents for direct reports. The Barangay Profiling System is responsible of providing residents and staff with an efficient and successful approach. They will be able to do work more quickly, and it also does away with the necessity for lots of employees. It will offer resident data based on profiles, responsibilities and permissions, administrators may monitor lab resources, user accounts, and reservation activities in real time from a single dashboard.

Barangay profiling promotes community participation and engagement in the development process. By involving residents in data collection, analysis, and decision-making, the system empowers communities to take ownership of their development and advocate for their needs.

Purpose and Description.

The main objective of the barangay profiling system is to provide stakeholders with a detailed understanding of the different aspects that define each barangay. The system aims to provide a comprehensive representation of the local environment by delving into various aspects of community life, such as population demographics, income distributions, access to basic services, housing conditions, environmental sustainability, and governance mechanisms.

The barangay profiling system is essential in promoting evidence-based decision-making, promoting community engagement, and encouraging diverse and sustainable development goals at the local level. By providing stakeholders with a thorough awareness of barangay realities and facilitating focused actions, the system helps to improve community well-being, adaptability, and growth.

Scope and Limitation of the Study.

The barangay profiling system is remaining a valuable tool for local development planning, resource allocation, and community empowerment,

emphasizing the importance of addressing challenges and implementing measures to maximize the benefits and effectiveness of profiling efforts in promoting inclusive and sustainable development at the barangay level. limited resources, including financial, human, and technological resources, may limit the scope and depth of barangay profiling activities, necessitating useful resource management and creative techniques to overcome limits and optimize the success of profiling activities.

STAKEHOLDER ANALYSIS

Identification of Stakeholders involved in the Project.

A number of parties are involved in the Barangay Profiling System: LGUs, including barangay councils, municipal or city governments, and provincial administrations, play a central role in barangay profiling. They are responsible for initiating, coordinating, and overseeing profiling activities within their respective jurisdictions non-governmental organizations (NGOs), civil society groups, and community-based organizations (CBOs) often collaborate with LGUs in barangay profiling initiatives. They help mobilize community participation, provide technical expertise, and advocate for community interests.

Roles and responsibilities of each stakeholder.

Stakeholders in the Barangay Profiling System each have certain roles and duties. The project must be approved by administrators, who also assign resources and make sure the initiative is in line with corporate objectives. Representatives from ethnic or indigenous groups within barangays ensure that the profiling process respects cultural sensitivities, traditional knowledge, and indigenous rights. NGOs, CBOs, and other community organizations collaborate with LGUs and barangay officials to support barangay profiling efforts. The Barangay Profiling System is developed, maintained, and improved to utilize profiling results to advocate for improvements, participate in decision-making processes, and contribute to community development initiatives.

Communication channels and frequency of Interactions.

Depending on their demands and duties, stakeholders in the Barangay Profiling System have different communication routes and frequency of interactions. Typically, project progress, budgetary allocations, and technological requirements are discussed in regular meetings or email exchanges between administrators.

Email communication is used for sharing updates, documents, and announcements related to barangay profiling activities. The frequency of email correspondence may vary depending on project milestones and deadlines. The system effective communication channels and regular interactions are essential components of the barangay profiling system. Stakeholders engage through various means such as meetings, email, social media, surveys, community consultations, and progress reports. These channels facilitate collaboration, information sharing, community engagement, and feedback collection throughout the profiling process. By utilizing a diverse range of communication methods and ensuring regular interactions, stakeholders can work together efficiently, address challenges effectively, and achieve the goals of inclusive and sustainable development within barangays.

CHAPTER II

REQUIREMENTS GATHERING:

Techniques used for gathering requirements (interviews, surveys, workshops, etc.)

In order to specify the features and functionalities of the Barangay Profiling System, gathering requirements for the barangay profiling system involves employing various techniques to understand the needs, preferences, and expectations of stakeholders. By using a combination of these methods, developers can gather comprehensive requirements that accurately reflect stakeholder needs and ensure the development of a system that effectively supports local governance and community development initiatives within barangays.

Summary of requirements elicited from stakeholders

The requirements for the Barangay Profiling System are gathered using a variety of methods. Comprehensive data collection, including demographics, socio-economic indicators, infrastructure, environmental factors, and governance structures. Stakeholders emphasize the importance of engaging community members in the profiling process, ensuring their active participation, ownership, and empowerment. This involves providing opportunities for input, feedback, and collaboration throughout the profiling cycle. By addressing these requirements, the profiling system can effectively support local governance, community development, and evidence-based decision-making within barangays.

Prioritization of requirements based on importance and feasibility

Stakeholders were asked to provide a range of requirements for the Barangay Profiling System to barangay officials, community members, government agencies, and development partners effectiveness in any Barangay's. Prioritizing requirements based on importance ensures that the barangay profiling system focuses on implementing features and functionalities that address critical needs, align with strategic priorities, and have a significant positive impact on community welfare and development. By focusing on implementing important requirements, the barangay profiling system can effectively support evidence-based decision-making, promote inclusive governance, and contribute to the overall well-being and development of barangays and their residents.

CHAPTER III

REQUIREMENTS ANALYSIS:

Detailed analysis of each requirement

Functional requirements:

Description:

- Functional requirements define the actions that the system should be able to perform, such as processing data, executing operations, or providing services to users.
- Features for user authentication, password management, and user profile customization.

Non-functional Requirements:

Performance

Description:

- The system should be capable of handling large volumes of data efficiently, with fast response times for data entry, retrieval, and analysis. It should also be scalable to accommodate future growth and increasing data.

Examples:

- The system can utilize scalable and efficient database technologies such as MySQL, PostgreSQL, or MongoDB to store profiling data. These databases are capable of handling large volumes of structured and unstructured data while providing fast read and write operations.

Security

Description:

- The system should encrypt sensitive data both in transit and at rest to prevent unauthorized access or interception.

Examples:

- This involves using strong encryption algorithms (e.g., AES) to encrypt data before transmitting it over networks and storing encrypted data in databases or file systems with robust encryption mechanisms.

Usability

Description:

- Optimizing system performance ensures that users experience fast response times and minimal latency when interacting with the system.

Examples:

- This involves optimizing database queries, minimizing page load times, and caching frequently accessed data to enhance overall system responsiveness.

Reliability

Description:

- Regularly backing up data and implementing robust backup and recovery procedures protects against data loss due to accidental deletion, corruption, or system failures.

Examples:

- The system administrator sets up automated backup schedules to regularly back up the profiling database and associated files.

Scalability

Description:

- Conduct performance testing and capacity planning exercises to identify potential bottlenecks and determine the system's scalability limits.

Examples:

- Use tools and techniques such as load testing, stress testing, and capacity modeling to validate scalability assumptions and optimize system performance.

Maintainability:

Description:

- Implementing automated testing practices, including unit tests, integration tests, and end-to-end tests, ensures the system's functionality remains intact after modifications or updates.

Examples:

- Using testing frameworks such as JUnit for Java or pytest for Python allows developers to write automated tests to validate system behavior and catch bugs early in the development cycle.

Compatibility:**Description:**

- The user interface of the barangay profiling system should adapt and adjust dynamically based on the screen size, resolution, and orientation of the device being used.

Examples:

- Menu options, buttons, and input fields should rearrange and resize to fit smaller screens on mobile devices, ensuring an optimal viewing and interaction experience.

Availability:**Description:**

- Develop a comprehensive disaster recovery plan outlining procedures for restoring service in the event of major disruptions, such as natural disasters, cyber-attacks, or data breaches.

Examples:

- This includes off-site backups, data replication, and failover sites to ensure business continuity and minimize downtime.

Compliance:**Description:**

- The system must implement robust security measures to safeguard sensitive data collected during profiling activities.

Examples:

- This includes encryption of data in transit and at rest, access controls to restrict access to authorized users, regular security audits and assessments, and incident response procedures to address security breaches or incidents

Use cases, user stories, or scenarios illustrating how the system will be used.

Use Cases:

Description: Use cases describe interactions between users (actors) and the system to achieve specific goals or tasks.

Format: Typically presented as a textual description or diagram, often using the "Actor-System-Action" format.

Example:

Use Case Name: Registration

Actor: Customer

Description: The customer registered to the said event.

User Stories:

Description: User stories capture high-level requirements from the perspective of end-users, focusing on the value delivered.

Format: Written in a simple, structured format: "As a [role], I want [goal], so that [benefit]."

Example:

User Story: As a customer, I want to be able to register to an application form for a summer job.

Scenarios:

Description: Scenarios describe specific sequences of actions or interactions between users and the system, often focusing on a particular use case or user story.

Format: Written as a narrative or step-by-step description of the user's actions and system responses.

Example:

Scenario:

Description: Ceejay is newly moved and wants to become a residence.

Steps:

Ceejay visits the barangay hall.

He applies the registration form.

Ceejay filling out the form.

After he filled out the forms, Ceejay became a official resident of the barangay.

Assumptions and constraints.:

Assumptions:

Description: Assumptions are beliefs or conditions that are taken for granted or considered to be true without conclusive evidence.

Purpose: Assumptions help fill in gaps in knowledge or information and provide a basis for decision-making.

Examples:

- "The system will be accessed by users with basic internet connectivity and modern web browsers."
- "The system will be hosted on a cloud infrastructure provided by a third-party vendor."
- "The development team will have access to necessary software tools and development environments."

Constraints:

Description: Constraints are limitations or restrictions that impose boundaries on the project or system design.

Purpose: Constraints define the parameters within which the system must operate and influence design choices and trade-offs.

Examples:

- "The project budget is limited to \$100,000, and no additional funding is available."
- "The system must be compatible with existing legacy systems and databases."
- "Regulatory requirements mandate that user data must be stored and transmitted securely in compliance with industry standards."

Types of Assumptions and Constraints:

Technical: Assumptions and constraints related to technology, infrastructure, or software platforms.

Resource: Assumptions and limits on project resources, such as budget, staff, and time.

Regulatory and Compliance: Assumptions and constraints imposed by legal or regulatory requirements, industry standards, or organizational policies.

Operational: Assumptions and constraints related to system operation, maintenance, and support.

Environmental: Assumptions and constraints stemming from the operating environment, such as geographical location or physical conditions.

Documentation:

- Assumptions and constraints should be documented explicitly in project documentation, such as the project charter, requirements specifications, or design documents.
- They should be reviewed and validated periodically throughout the project lifecycle to ensure their continued relevance and accuracy.
- Any changes to assumptions or constraints should be communicated to relevant stakeholders and incorporated into project planning and execution as necessary.